UNITY EDITOR EXTENSION

Fracture Tool Free

Real-time mesh fracturing for Unity: Voronoi and Simple fracture with a custom inspector. Free edition.

Version 1.0 | Unity 2022.3 LTS or newer | Built-in Render Pipeline demo (runtime code is pipeline independent)

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1. Overview

Fracture Tool breaks any mesh into physically simulated fragments at runtime. Add a single component, `FractureController`, pick a fracture mode and the object can be broken by a click or from your own scripts.

- **Two fracture modes:** Voronoi and Simple.
- **Physics settings:** mass, drag, gravity scale, bounciness, friction and explosion strength.
- **Interior material:** cut faces get their own material, so the inside of a log looks different from its bark.
- **Custom inspector:** preview the cut layout in the Scene view, both in Edit and Play Mode.
- **Clean runtime:** fragments release their generated meshes and material instances when destroyed.

All code lives in the `FractureTool` namespace and compiles into its own assemblies (`FractureTool` and `FractureTool.Editor`).

2. Quick Start

1. Import the package and open `Assets/FractureTool/Scenes/Demo.unity` to see every feature in action.
2. Select the object you want to break. It needs a **MeshFilter**, a **MeshRenderer** and a **Collider**.
3. If the object comes from an imported model, enable **Read/Write** in the model import settings. The inspector shows a warning and a one-click fix if it is disabled.
4. Add the **Fracture Controller** component (*Add Component* > *Fracture Controller*).
5. Choose a **Fracture Mode** and, optionally, an **Interior Material** for the cut faces.

6. Press **Preview Fracture** to see the cut layout, then enter Play Mode and click the object (or press **Fracture Object** in the inspector).

> The mode specific component (for example `VoronoiGenerator`) is added and removed automatically when you change the mode. Its settings are shown inside the Fracture Controller inspector.

3. Fracture Modes

Mode	Result	Best for	Main settings
Voronoi	Organic, irregular cells built from random seed points inside the object.	Rocks, walls, pottery, generic debris.	Number of Points (3 to 200).
Simple	One horizontal cut through the center.	Logs, pillars, quick splits.	None.

4. Fracture Controller Reference

Main Configuration

Fracture Mode	Algorithm used to break the object.
Preview Mode	When enabled, clicks and `TriggerFracture()` only show the cut layout instead of breaking the object.

Materials

Interior Material	Material applied to the new cut faces. If empty, the object's own material is used.
Fragment Material	Replaces the outer material of the fragments. If empty, the original material is kept.

General Physics

Apply Physics	Adds a Rigidbody to every fragment.
Destroy Original	Destroys the source object after breaking it. If disabled, it is hidden instead.
Fragment Separation	Small offset that pushes fragments apart when they are created.
Use Explosion Force	Applies an outward impulse and spin. When disabled, fragments simply collapse under gravity.
Global Explosion Force	Base strength of the impulse, multiplied by the Custom Physics force multiplier.
Auto Destroy / Destroy After	Removes fragments after the given number of seconds.

Actions

Preview Fracture	Draws the cut layout in the Scene view (seed points, cut planes or grid). Works in Edit and Play Mode.
Fracture Object	Breaks the object. Available in Play Mode only, since fragments are physics objects.

5. Scripting API

```
using FractureTool;

FractureController fc = GetComponent<FractureController>();

fc.Fracture();              // Break now (Play Mode). Ignores Preview Mode.
fc.Preview();               // Show the cut layout without breaking.
fc.TriggerFracture();       // Preview or break depending on fc.previewMode.
bool broken = fc.HasFractured;

fc.SetFractureMode(FractureController.FractureMode.Simple);
fc.globalExplosionForce = 4f;
fc.SetCustomPhysicsPreset("bouncy");   // lightweight, heavy, bouncy, default
fc.GetVoronoiComponent().pointCount = 20;
```

`VoronoiGenerator` can also be used on its own through `Fracture(GameObject)`, and `FractureUtility` exposes the slicing helpers used internally.

6. Demo Scene

Open `Assets/FractureTool/Scenes/Demo.unity` and press Play. Everything in the scene is built from Unity primitives and procedurally generated textures, so no extra packages are needed.

WASD + Mouse	Move and look around.
Left Click	Break the object under the crosshair.
E	Start a round in the shooting range. Hold Left Click to fire.
R	Reset the scene.
H / Esc	Hide the help / release the cursor.

The front row shows Voronoi and Simple fracture. Behind them, a fluted marble column (about 4,000 triangles) and a faceted boulder show Voronoi fracture on detailed meshes, and the corners hold the ready-made prefabs from `Assets/FractureTool/Prefabs`.

Shooting Range

Walk through the north door into the shooting range and press **E** to start a 60 second round. Targets pop up on three shelves, some of them moving, and physical projectiles break them through the public API (`FractureController.Fracture()`). Wood, stone, glass, ceramic plates and marble columns give different scores; the best score is saved. The scripts in `Demo/Scripts` (`ShooterGame`, `ShooterWeapon`, `ShooterProjectile`, `ShooterTarget`) are a compact example of how to hook the tool into gameplay. The demo uses the Built-in Render Pipeline *Standard* shader. On URP or HDRP, convert the demo materials with the Render Pipeline Converter; the fracture code itself does not depend on the render pipeline.

7. Performance Tips

- Fracturing happens on the main thread. Keep fragment counts reasonable for objects that break during gameplay: 8 to 30 fragments is usually enough.
- Low poly meshes cut faster. Use a simplified mesh for objects that will be broken.
- Keep **Auto Destroy** enabled so fragments do not accumulate. Destroyed fragments free their generated meshes and material instances.
- Voronoi cost grows with the number of points.
- Fragments use convex Mesh Colliders. For very large numbers of small debris pieces, consider lowering **Destroy After**.

8. Troubleshooting

Nothing happens in a build	Enable **Read/Write** on the model. The mesh must be readable on the CPU to be cut.
Clicking does not break	The object needs a Collider, and **Preview Mode** must be disabled for clicks to break it.
"produced no fragments" warning	The cut region does not intersect the mesh. Check that the collider matches the visible mesh.
Pink demo materials	The project uses URP or HDRP. Convert the materials in `Demo/Materials`.

9. Package Contents

```
Assets/FractureTool/
  Scripts/                      Runtime code (FractureController)
    Fracture/                   Fracture modes and helpers
      Editor/                   Custom inspectors (editor only)
    Player/                     Simple first person controller used by the demo
  ExternalLibraries/EzySlice/   Mesh slicing library (MIT)
  Demo/                         Demo and mini game scripts, materials, textures, meshes
  Prefabs/                      Ready-made crate, rock, column and boulder
  Scenes/Demo.unity             Demo scene
  Documentation/                This document
  Third-Party Notices.txt
```

10. Third-Party Notices

Fracture Tool includes **EzySlice** by David Arayan, licensed under the MIT License. The namespace was changed to `FractureTool.EzySlice` so it does not conflict with other copies of the library in your project. See `Third-Party Notices.txt` for the full license text.

11. Fracture Tool Pro

The Pro edition adds Radial, Random Planes and Grid fracture, Wood / Stone / Glass / Metal material presets, five UV correction methods and trigger and collision fractures. It uses the same GUIDs as this edition: import it over Fracture Tool Free and your scenes keep working.

12. Support

Questions, bug reports or feature requests: **studiosnaxo@gmail.com**

Please include your Unity version, the fracture mode you are using and, if possible, the Console output or a small reproduction scene.